

Early graphics drawing programs held very simple information for vector images — not much more than the coordinates for the various primitives and colours for the areas formed by the various shapes and line.

In modern systems, of course, there is very much more information held for each element in the image. However, the principle remains the same. There are two important issues with vector images: they can very easily be edited, as the coordinates are simply numbers, and there is effectively no limit to the amount of detail that can be held — the only limit being the maximum values available for the coordinate system. When they come to be used, it is normal that a vector graphic is translated into some form of raster image, either directly onto a screen, or into a file format. This process is called rendering, and when a vector image is rendered to create a raster image file, a resolution has to be specified for the final image. It may well be the case that, if a low resolution is required, then some fine details in the vector original will not be visible; the higher the resolution used, the more such detail can be represented.

The application to animation is clear: as the coordinates are simply numbers, changing in various ways the coordinates of the points defining objects becomes easy. This can be done programmatically, according to a large variety of different rules. As long as these rules can be programmed into the system, objects in a scene — we are now assuming 3D scenes in which there are three coordinates for each point rather than two — can be animated one frame at a time, and each frame rendered automatically.

This is what is done by 3D software such as Autodesk's 3ds MAX or Maya, or LightWave 3D from NewTek, to name just a few: these programs enable the creation of 3D models of scenes and objects. These can be useful in their own right. Modern systems allow for highly realistic image creation, and just still images maybe all that is needed for many purposes — such as an architect's drawing of a proposed building or an engineer's gear-wheel. One point about rendering: pre-rendered scenes are rendered with the modelling software, such as MAX. You might do this for an architectural fly-through or film animation (*South Park*, for example, is created using MAX). The alternative is real-time rendering; here the scene is created using modelling software, but the rendering is done by perhaps very different software, in real-time. This is the case with games.

The way in which objects are constructed in a scene in software such as 3ds MAX can be considered as an extension of the methods of vector graphics. The most basic elements are

points (vertices), each with three coordinates, and the most basic surface feature is a triangle, defined by three such points. Although there are other ways of describing surfaces, and we'll touch on one of these later, we can consider the surfaces of all objects to consist of a patchwork of triangles. The more complex the surface, naturally the greater the number of triangles that will be needed.

You can create a complex object by building up one triangle after another — and sometimes getting down to low-level detail like this is necessary — but it is usual to construct models from basic primitives. One example: consider a gear-wheel; you could create a simple 2d (vector) line drawing of the cross section of the gear, either in the modelling software itself or import it from some CAD or other 2D drawing software. The shape is then extruded — it is given a third dimension by the software, stretched up, one could say, to give the 3D gear-wheel. In that automatic process, all the necessary vertices and triangles will be created by the software.

Modelling programs also allow for the creation of simple primitives directly —



Inside the basilica, again before and after rendering. Notice how, without proper lighting techniques in the edit window, the lighting appears flat and without character